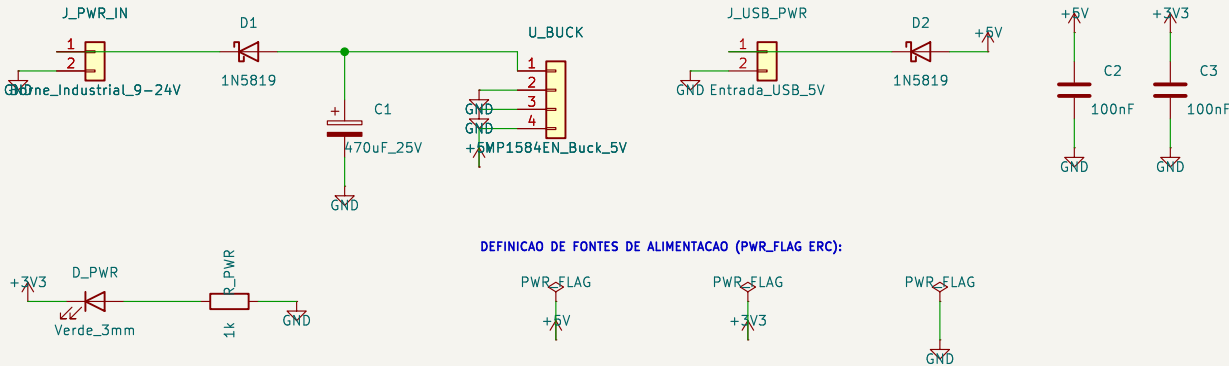


SISTEMA DE METROLOGIA LEAN TECH – GATEWAY CONCENTRADOR INDUSTRIAL (LORA TO MQTT)

Ponte de Borda LoRa 915 MHz (LLCC68) para Wi-Fi / Ethernet W5500 com Ingestao MQTT para o Backend Laravel

BLOCO 1: ALIMENTACAO INDUSTRIAL & GERENCIAMENTO DE ENERGIA (9V–24V DC / USB 5V)

Entrada por Borne 9–24V com protecao reversa e Step-Down MP1584EN + Entrada auxiliar USB–C 5V



DEFINICAO DE FONTES DE ALIMENTACAO (PWR_FLAG ERC):

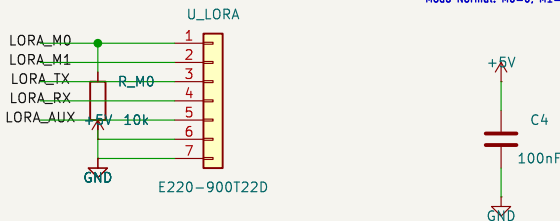
BLOCO 3: COMUNICACAO SEM FIO LORA LONGO ALCANCE (CDEBYTE E220–900T22D)

Chip Semtech LLCC68 | Frequencia 915 MHz | Potencia 22 dBm (160 mW) | Alcance ate 5 km | Interface UART

PINAGEM EBYTE E220–900T22D (PADRAO DIP 7 PINOS):

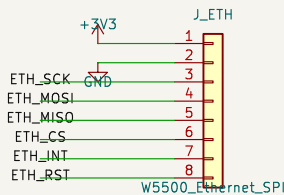
1: M0 | 2: M1 | 3: RXD | 4: TXD | 5: AUX | 6: VCC (5V) | 7: GND

Modo Normal: M0=0, M1=0 (Transmissao transparente serial automatica)



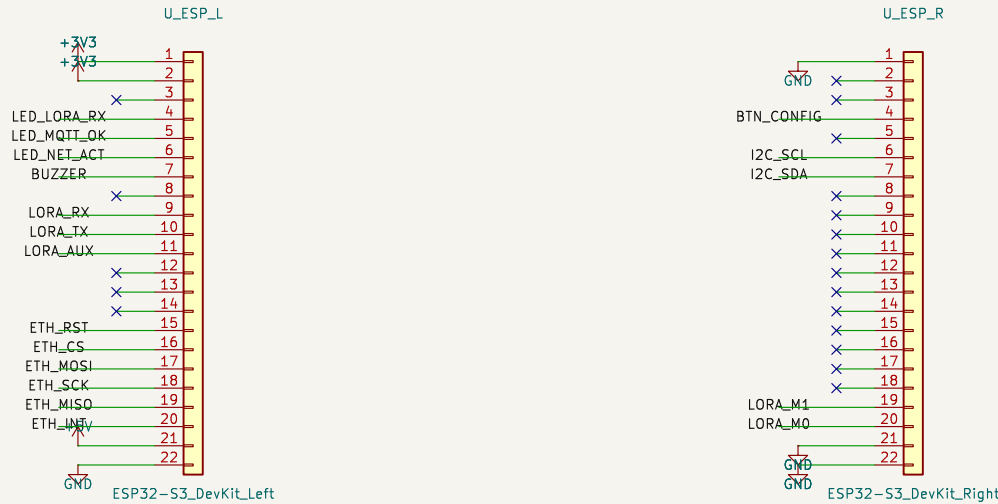
BLOCO 4: INTERFACE ETHERNET CABEADA (SOCKET MODULO W5500 SPI)

Conexao cabeada RJ45 10/100 Mbps com protocolo TCP/IP embarcado em hardware | Ideal para chão de fábrica



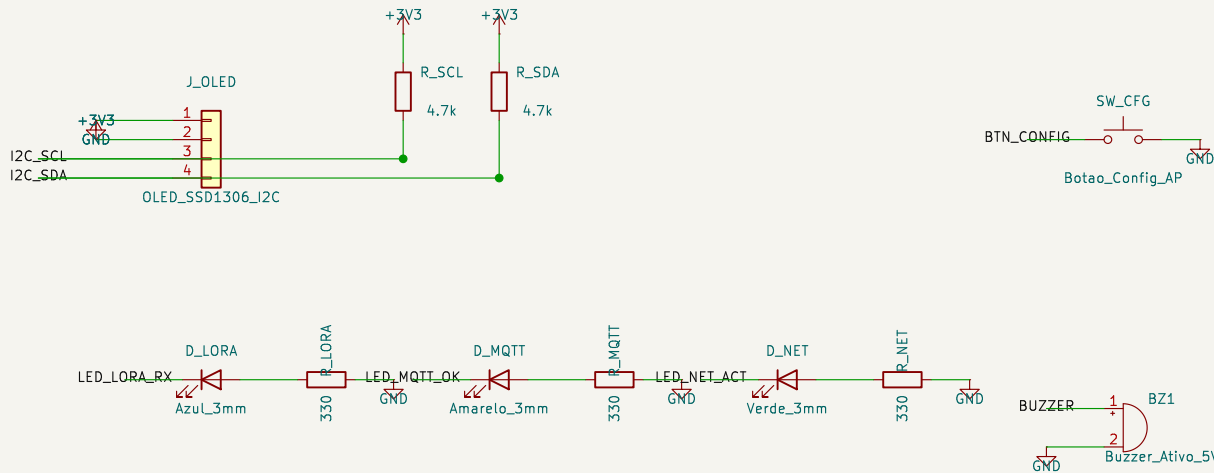
BLOCO 2: MICROCONTROLADOR PRINCIPAL (ESP32–S3 DEVKITC–1 44 PINOS DUAL ROW)

Soquetes femae duplos 1x22 pinos | Wi-Fi 802.11 b/g/n + BLE 5.0 | Dual Core 240 MHz LX7



BLOCO 5: DIAGNOSTICO VISUAL & IHM LOCAL (DISPLAY OLED 0.96, LEDS DE STATUS E BUZZER)

Display I2C para IP, RSSI LoRa e status MQTT | LEDs de atividade em tempo real | Botao de Modo AP



Transceptor: CDEBYTE E220–900T22D (LLCC68 915 MHz 22dBm)
Padrao de Projeto: IEEE 315 / IEC 61082–1 / IPC–2221B
Certificacao: 0 Violacoes ERC / 0 Violacoes DRC

Lean Tech Metrology & Industrial IoT Solutions

Sheet: /
File: amemiya_gateway.kicad_sch

Title: Amemiya Industrial IoT – Gateway Concentrador LoRa/MQTT v1.0

Size: A3 Date: 2026–09–28

KiCad E.D.A. 9.0.7

Rev: v1.0

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